

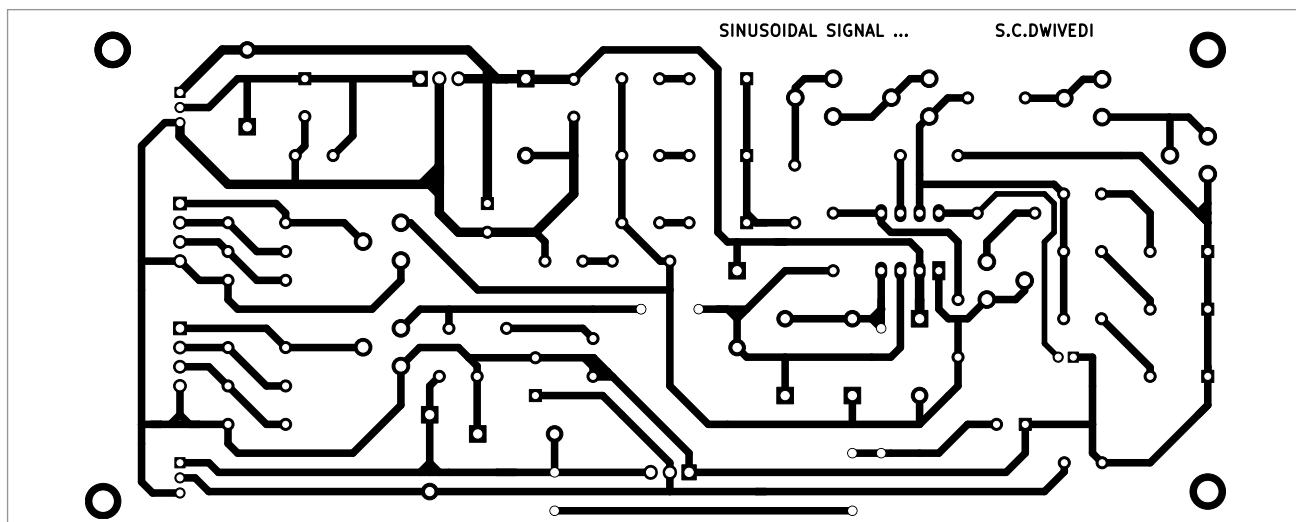


Fig. 2: Actual-size, single-side PCB

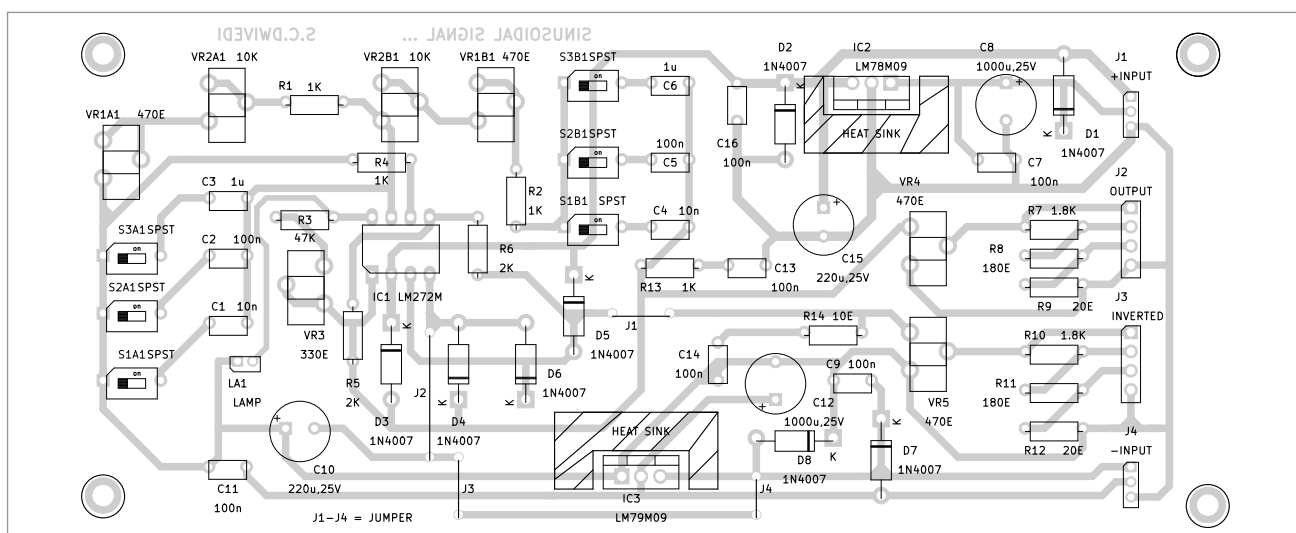


Fig. 3: Component layout of the PCB

IC1B across terminal J3. The output OUT2 is also produced from the first oscillator IC1A, but it is inverted. The amplitude of these outputs is adjusted with potmeter VR5. OUT2:10 is divided by 10 of OUT2, while OUT2:100 is divided by 100 of OUT2.

The snubber circuits at the outputs of op-amps are not obligatory. But in case of need, we can connect in series a resistor of 1- to 3-ohm and capacitors of 47 to 220nF with each op-amp, depending on expected loads. To stabilise the amplitude of IC1A, we may use lamps rated at 6V/20mA, 6V/40mA, 12V/20mA, etc.

The circuit can be used at home,

in a lab, or in the field, because it can work from battery ($2 \times 4.5V$, $2 \times 6V$, or $2 \times 9V$). It needs a simple adjustment of the amplitude of the oscillator built around IC1A to work properly.

Construction and testing

An actual-size, single-side PCB for the sinusoidal signal generator is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on a PCB, connect the power supply across terminals J1 and J4 in the PCB. Fix potentiometers VR1 through VR5 on the front side of the cabinet so that you can adjust them to get

the required frequencies.

EFY Notes. 1. The simultaneous adjustment of the output voltages can be made with a stereo potentiometer for VR1 (VR1A1, VR1B1) and VR2 (VR2A1, VR2B1)

2. The output voltages can be adjusted more precisely with individual potentiometers, say, VR4 and VR5. **EFY**

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